Alex Berns

Email: berns.alexander@gmx.net
Website: alexberns.github.io/berns
Citizenship: Germany

Experience

TSMC, Dresden, Germany
CMP Equipment Engineer (starting April 2027) — 2027–

SB Intuitions (SoftBank Group), Tokyo, Japan
Research Intern, Responsible AI Team — 2026
AI Safety research on Japan's largest GPU cluster (until Dec 2026)

Feluwa Pumpen GmbH, Mürlenbach, Germany
Dual Study, Electrical Engineering (R&D) — 2019–2024
Machine-learning model for fault diagnosis of double-hose membrane pumps
Mechatronics Technician (apprenticeship) — 2019–2023
Switchgear, frequency converters & motors up to 1.2 MW; plant maintenance
Assembly of control panels & connection of sensor equipment

Trier University of Applied Sciences, Germany
Teaching Assistant, Undergraduate Mathematics — 2021–2022

German–Chinese Society, Trier, Germany
Chinese Language & Cultural Assistant — 2020–2023
Translation, interpreting & cultural exchange

Education

Tohoku University, Sendai, Japan
M.Sc. Computer Science — 2024–2027
Advisor: Minoru Kuribayashi; Ito Foundation Full Scholarship (JPY 3.6M)
Also admitted to the Technical University of Munich (TUM)

Trier University of Applied Sciences, Germany
B.Eng. Electrical Engineering — 2019–2024
GPA 3.4/4.0; thesis: ML for condition monitoring of membrane pumps

Berufsbildende Schule Wittlich, Germany
Certified Mechatronics Technician (IHK) — 2019–2023
Recognized German dual-vocational qualification

Westlake Boys High School, New Zealand
Cambridge International A-Levels — 2015–2019
A*A*A — Physics, German, Mathematics

Publications

Toxicity Lost in Translation: A Toxicity-Preserving Framework for LLM Safety Benchmarking
Berns, Huy, Pride, Wataoka — AAAI 2027, Montréal, Canada

Layer-Wise Weight Statistics for Node Classification and Defense of Federated Large Language Models
Berns, Akai, Kuribayashi — APSIPA ASC 2025, Singapore

Selected Awards

FunkenSprüher Innovation Award for academic thesis — 750 EUR — 2024

German National Chinese Language Competition — 2nd place — 2023

AFS Exchange Scholarship (New Zealand Government) — six months with a Chinese host family, attending high school in Yantai, Shandong — 2018

Languages

German (native), English (native), Chinese (Mandarin, HSK 6), Japanese (JLPT N3)

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